2. Write the formula for the following coordination compounds.
   a) potassiumhexacyanidoferrate(II)
   b) petacarbonyliron(0)
   c) pentaamminenitrito $-\kappa-N$ -cobalt(III)ion
   d) hexaamminecobalt(III)sulphate
   e) sodiumtetrafluoridodihydroxidochromate(III)
3. Arrange the following in order of increasing molar conductivity
   i) $Mg[Cr(NH_3)(Cl)_5]$ ii) $[Cr(NH_3)_5Cl]_3[CoF_6]_2$
   iii) $[Cr(NH_3)_3Cl_3]$
4. Ni^{2+} is identified using alcoholic solution of dimethyl glyoxime. Write the structural formula for the rosy red precipitate of a complex formed in the reaction.
5. $[CuCl_4]^{2-}$ exists while $[CuI_4]^{2-}$ does not exist why?
6. Calculate the ratio of $\frac{[Ag^+]}{[Ag(NH_3)_2]^+}$ in 0.2 M solution of NH_3. If the stability constant for the complex $[Ag(NH_3)_2]^+$ is 1.7×10^7
7. Give an example of coordination compound used in medicine and two examples of biologically important coordination compounds.
8. Based on VB theory explain why $[Cr(NH_3)_6]^{3+}$ is paramagnetic, while $[Ni(CN)_4]^{2-}$ is diamagnetic.
9. Draw all possible geometrical isomers of the complex $[Co(en)_2Cl_2]^+$ and identify the optically active isomer.
10. $[Ti(H_2O)_6]^{3+}$ is coloured, while $[Sc(H_2O)_6]^{3+}$ is colourless- explain.
11. Give an example for complex of the type $[Ma_2b_2c_2]$ where a, b, c are monodentate ligands and give the possible isomers.
12. Give one test to differentiate $[Co(NH_3)_5Cl]SO_4$ and $[Co(NH_3)_5SO_4]Cl$.
13. In an octahedral crystal field, draw the figure to show splitting of d orbitals.
14. What is linkage isomerism? Explain with an example.
15. Write briefly about the applications of coordination compounds in volumetric analysis.
16. Classify the following ligand based on the number of donor atoms.
    a) NH_3 b) en c) ox^{2-} d) triaminotriethylamine e) pyridine

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